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CAMPUS NAVIGATION AND SHORTEST PATH FINDER

ABSTRACT

Navigating between buildings in a large college campus can be difficult for new students, visitors, and staff, particularly when they are unfamiliar with the available walking routes. The proposed Campus Navigation and Shortest Path Finder aims to help users identify a suitable route between selected campus locations by applying concepts from Advanced Data Structures and Algorithms. The project will use the provided Anna University Main Campus map as a reference to identify important places such as academic departments, the central library, hostels, canteens, parking areas, and entrances. These locations will be represented as vertices in a graph, while walkable paths connecting them will be represented as edges. Each edge will be assigned a distance weight based on measured or clearly documented estimated distances.

The system will allow a user to select a starting location and a destination and calculate a route using Dijkstra's algorithm for weighted paths. Breadth-First Search (BFS) and Depth-First Search (DFS) will be included to demonstrate graph traversal, while the Floyd–Warshall algorithm will be used to calculate shortest distances between all pairs of selected locations. Prim's or Kruskal's algorithm may also be implemented to demonstrate how a minimum spanning tree can connect locations with minimum total edge weight. A user interface will display the selected locations, route, and calculated distance, and the project will compare the algorithms' outputs and execution times on the same campus graph.

The application is planned to be developed using Python for graph construction and algorithm implementation, with Flask, HTML, CSS, and JavaScript for the interface. The project will be tested with different source–destination pairs and checked against expected results. The final outcome will be a practical campus-routing prototype and an educational demonstration of graph representations, shortest-path algorithms, heaps, and minimum spanning trees. The accuracy of route results will depend on the completeness of the campus graph and the reliability of the distance data.

Keywords: Campus Navigation, Graph Algorithms, Dijkstra's Algorithm, BFS, DFS, Floyd–Warshall, Minimum Spanning Tree, Shortest Path.